NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Evaluation and Programme Management cont…



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CONCEPTS COVERED

- Cash-flow forecasting
- Detailed steps of C-B Analysis
- Categories of costs and benefits



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Cash-flow forecasting

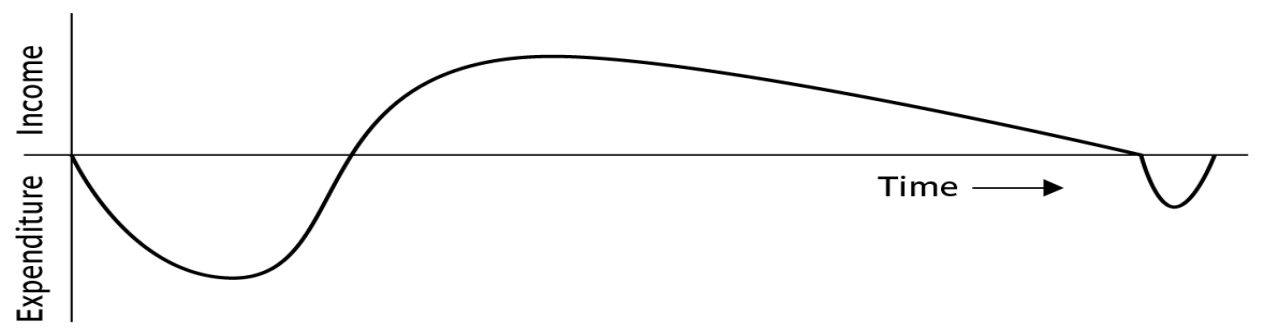
- Like C-B analysis, it is also important to produce a cash-flow forecast, which indicates when expenditure and income will take place.
- The development of the project will incur costs. We need to spend money such as purchasing equipments, staff payment during project development.
- These expenses cannot wait until income is received from the project.
- When the system or product is released it will generate income that gradually pays off costs.
- Also, the timing of costs and income for a product of system needs to be estimated.





Product/system life cycle cash flows

- We need to know that, whether we can fund these expenses either from the own resources or by borrowing.
- So, a forecast is needed.



Typical product life-cycle cash-flow

Cost-benefit analysis – Detailed Steps

1. Identify the costs and benefits pertaining to the project
2. Categorize the various costs and benefits
3. Select a cost-benefit evaluation technique
4. Interpret the results of the analysis
5. Take appropriate action.

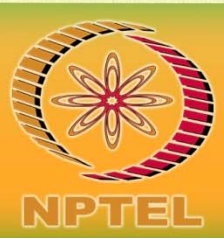




Identify the costs and benefits pertaining to the project

- Certain costs and benefits are easily identifiable, e.g. direct costs, such as the price of a computer etc. can be easily identified.
- Direct benefits often relate one-to-one to direct costs, especially savings from reducing costs in the activity in question.
- Some direct costs and benefits, may not be well defined, since they represent estimated costs or benefits that have some uncertainty, e.g. cost reserved for bad debt.
- A category of costs or benefits that is not easily identifiable is opportunity costs and benefits. These are the costs or benefits forgone by selecting one alternative over another.





Categorize various costs and benefits for analysis

- Tangible or Intangible
- Direct or indirect
- Fixed or variable





Categorize various costs and benefits for analysis

Tangible costs.

- Tangibility refers to the ease with which costs or benefits can be measured.
- An outlay of cash for a specific item or activity is referred to as a tangible cost.
- Examples: purchase of hardware or software, personnel training, and employee salaries.

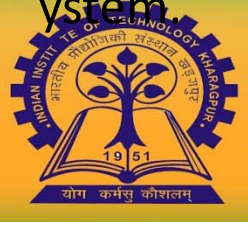


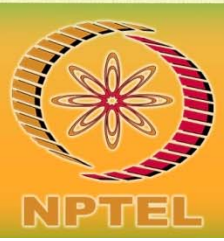


Categorize various costs and benefits for analysis cont...

Intangible costs

- Costs that are known to exist but whose financial value cannot be accurately measured.
- Example: employee morale problems caused by a new system or lowered company Image.
- In some cases, they are easy to Identify but difficult to measure, e.g. the cost of breakdown of an online system during banking hours will cause the bank to loose deposits and waste human resources. But, by how much? Difficult to measure.
- In some cases, these costs may be difficult even to identity, e.g. improvement in customer satisfaction due to online systems. ystem.





Categorize various costs and benefits for analysis cont...

Tangible benefits

- Benefits which can be measured/quantified.
- Examples: Benefits due to completing jobs in fewer hours or producing reports with no errors.



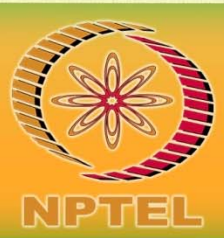


Categorize various costs and benefits for analysis cont...

Intangible benefits

- Benefits which cannot be easily measured / quantified.
- Example: More satisfied customers or an improved corporate image.
- Both tangible and intangible costs and benefits, should be considered in the evaluation process.
- Management often tends to deal irrationally with intangibles by ignoring them, which should not be.





Categorize various costs and benefits for analysis cont...

Direct Costs

- Costs those with which a dollar figure can be directly associated in a project.
- Easy to quantify in monetary terms.
- For example, the purchase of a box of diskettes for $35 is a *direct cost* because we can associate the diskettes with the dollars expended.
- Other examples: Development cost, setup cost, operational cost .





Categorize various costs and benefits for analysis cont...

Direct benefits

- Benefits which can be specifically attributable to the given project.
- Example: A new online system that can handle 25 percent more transactions per day.





Categorize various costs and benefits for analysis cont...

Indirect costs

- These costs are the results of operations that are not directly associated with a given system or activity. They are often referred to as overhead.
- Example: Insurance, maintenance, protection of the computer center from heat and fire, lighting, and air conditioning.
- But, it is difficult to determine the proportion of each, attributable to a specific activity.





Categorize various costs and benefits for analysis cont...

Indirect benefits

- The benefits which are realized as a by-product of another activity or system.

- Examples: benefits due to selling of fertilizers in a steel plant, selling of empty oil containers (tin boxes), etc.





Categorize various costs and benefits for analysis cont...

Fixed Costs

- Some costs are constant, regardless of how well a system is used. They are called *fixed costs*.
- Once encountered, they will not recur.
- Examples: Straight-line depreciation of hardware, exempt employee salaries, and insurance etc.





Categorize various costs and benefits for analysis cont...

Variable costs

- These costs are incurred on a regular (weekly, monthly) basis.
- They are usually proportional to work volume and continue as long as the system is in operation.
- Example: Costs of computer forms which vary in proportion to the amount of processing or the length of the reports required.





Categorize various costs and benefits for analysis cont...

Fixed benefits

- These benefits are also constant and do not change.
- Example: Benefits due to the decrease in the number of personnel by 20 percent resulting from the use of a new computer.
- The benefit of personnel savings may recur every month.





Categorize various costs and benefits for analysis cont...

Variable Benefits

- On the other hand, these benefits are realized on a regular basis.
- Example: Consider a safe deposit tracking system that saves 20 minutes preparing customer notices compared with the manual system. The amount of time saved varies with the number of notices produced.





Summary

- Discussed the steps for carrying out C-B Analysis.
- Learnt the different types of costs and benefits with suitable examples.
- These costs and benefits have to be considered properly in C-B analysis.





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